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Ownership Similarity and Investor Identity: A Quantitative Study on Family Office and Private Equity Deals

23.10.2025 | Stefano Sima

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Introduction

01

Motivations & Research Gaps

The rise of Family Offices



FOs are growing rapidly and adopting **PE-style investment strategies**, but research on them is still **limited and mostly qualitative**.

Few quantitative comparisons with PEs



While **PEs are well-studied**, **large-scale, data-driven comparisons** between FOs and PEs are **rare and underdeveloped**.

Ownership similarity remains underexplored

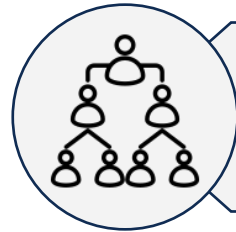


The idea of **ownership similarity** has been **theorized** (Bettinazzi *et al.*, 2020) but **not quantitatively tested** for its **impact on firm performance**.

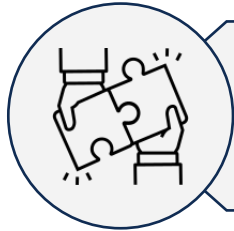
Ownership Similarity & Implications

Reflects how much investor and target “resemble” each other in **values, decision-making, and long-term orientation.**

Relevance in Family Firms



Family Businesses value **trust, continuity,** and **legacy preservation.**



Family Offices often share similar **governance principles**, easing integration and collaboration.



Private Equity firms, focused on **financial returns** and **exit strategies**, may face more **alignment challenges.**

Implications for Investment Dynamics



High ownership similarity → smoother **deal selection, evaluation,** and **negotiation.**



Facilitates **trust, information interpretation,** and **strategic alignment.**



Enhances **post-deal cooperation** and reduces **cultural or governance conflicts.**

Objectives & Hypotheses

02

Objectives & Hypotheses

- Compare **Family Offices** and **Private Equity** firms in investment patterns and outcomes.
- Test whether **ownership similarity** influences post-deal performance.
- Explore differences in **deal structure** (minority vs. majority stakes).

HYPOTHESIS 1

Family Offices are more likely than Private Equity firms to invest in family-owned businesses.

HYPOTHESIS 2

Firms backed by investors with high ownership similarity (such as Family Offices investing in family businesses) perform better post-deal than firms backed by investors with low ownership similarity (such as Private Equity firms investing in family businesses).

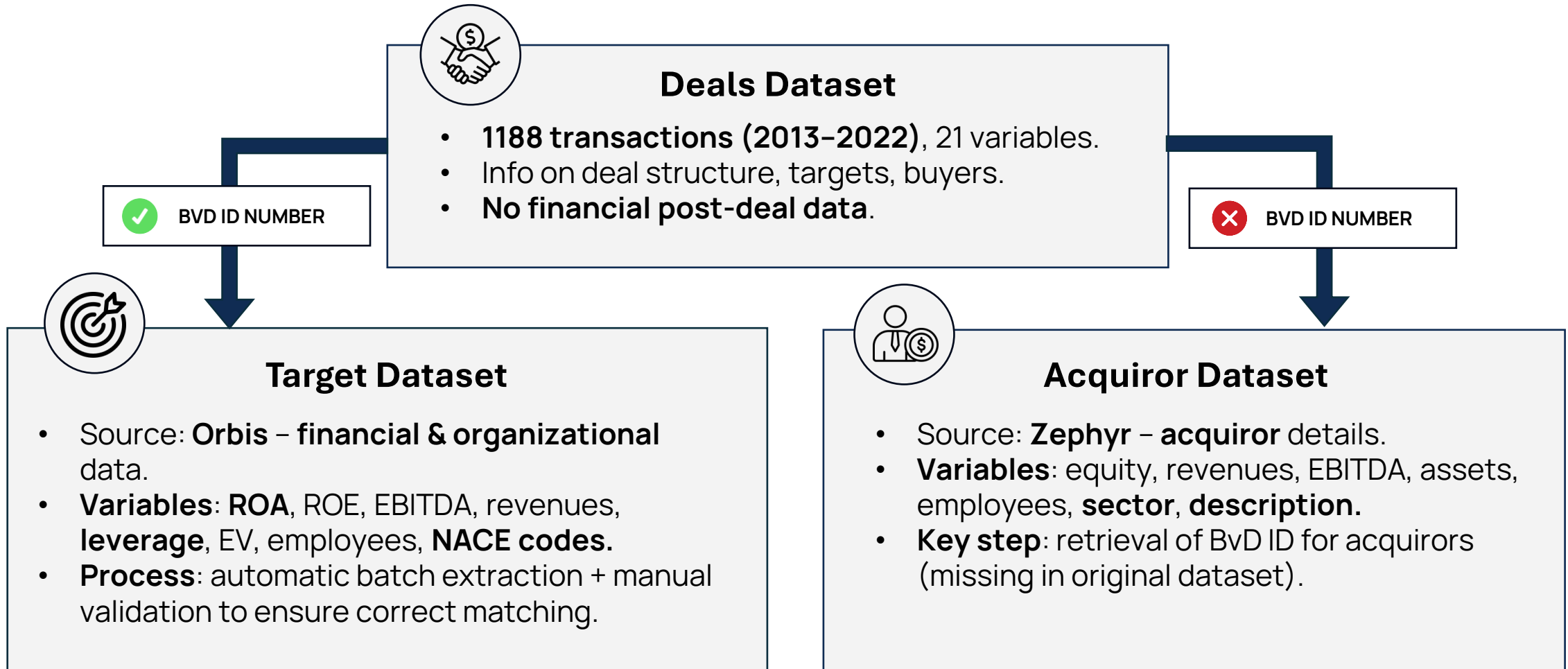
HYPOTHESIS 3

Family Offices are more likely than Private Equity firms to engage in minority-type deals, while Private Equity firms more frequently complete majority-type deals.

Methodology

03

Dataset Construction



Investor Classification

Objective



- Identify **Family Offices** and **Private Equity** firms.
- Crucial to test whether investor identity affects deal choices & outcomes.

Challenges

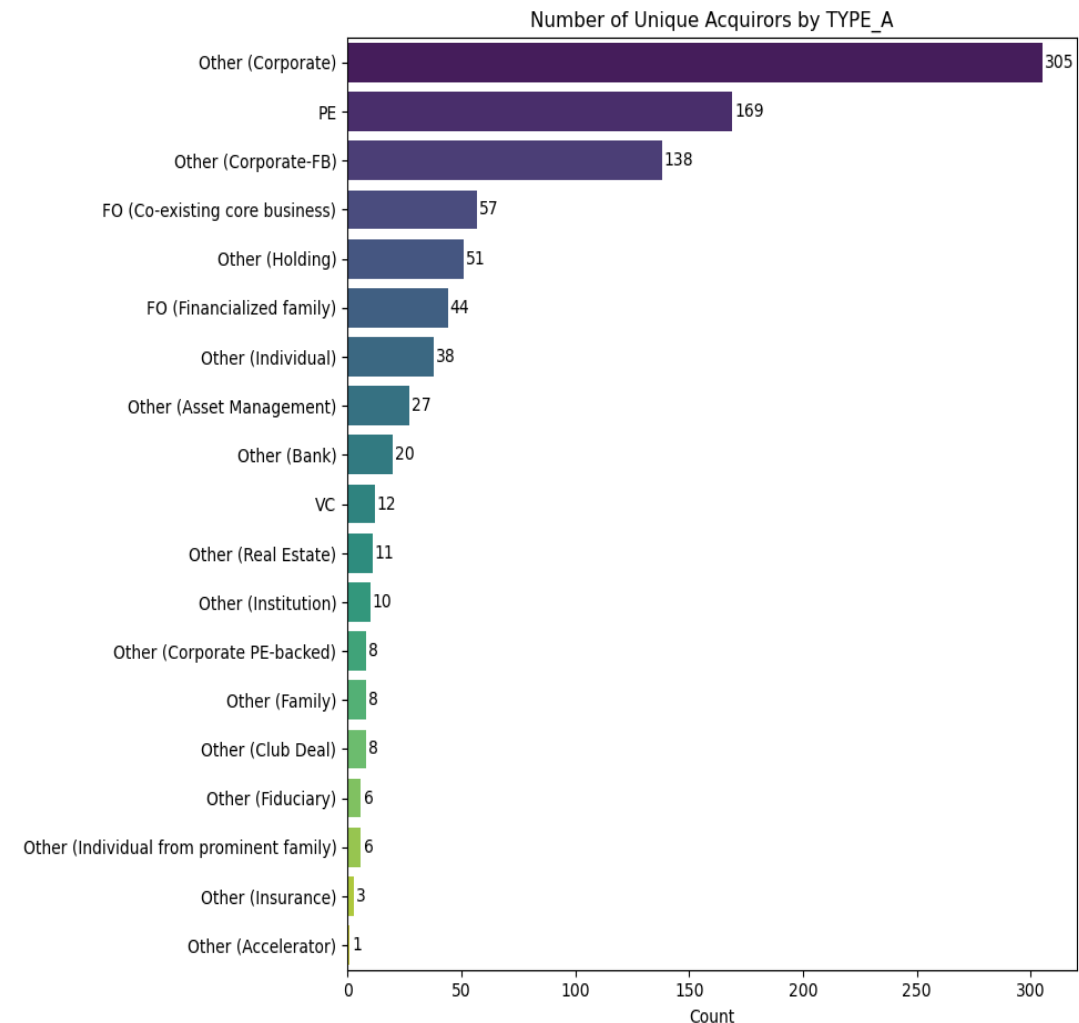


- **No official registry of FOs in Italy.**

Procedure



1. **Regulatory check** → Consob & Bank of Italy registers (regulated PE firms).
2. **Ownership & governance** → Zephyr + Orbis + company websites (company description, family ties, governance, declared strategy).
3. **Cross-verification** → financial news outlets (BeBeez, Il Sole 24 Ore, Milano Finanza, Dealflower).



Data Preparation



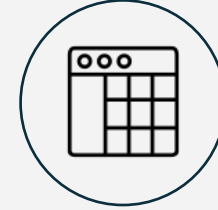
Cleaning & Integration

- Merged the 3 dataset: Deals, Targes and Acquirors.
- Removed duplicates & reorganized multi-buyer/target deals.
- Dropped incomplete/generic entries.



Standardization & Adjustments

- Missing values standardized.
- Winsorization of extreme outliers.
- Log-transformation: firm size, leverage.
- ROA and Δ ROA smoothed.
- Creation of new variables.



Final Dataset

- 314 deals (208 PE, 106 FO) → Focus only on Family Offices and Private Equity firms.
- Variables aligned at $t-1$, t , $t+1$ to track pre/post-deal changes.



Econometric Approach

H1 – Investor Preference for Family-Owned Targets



Chi-squared test → association FO vs. PE × family ownership



Logistic regression → probability of FO vs PE investment
(controls: size, leverage, profitability, sector)

H2 – Ownership Similarity & Post-Deal Performance



Δ ROA (t+1 – t-1) as performance measure



Robust regression (Tukey biweight) → handle outliers & non-normality



Heckman two-step → correct for selection bias in target choice

H3 – Deal Type and Investor Identity



Chi-squared test → compare FO vs PE control preferences



Logistic regression excluded due to **sample imbalance** (majority ≈ 70% of all deals)

Analysis & Results

04

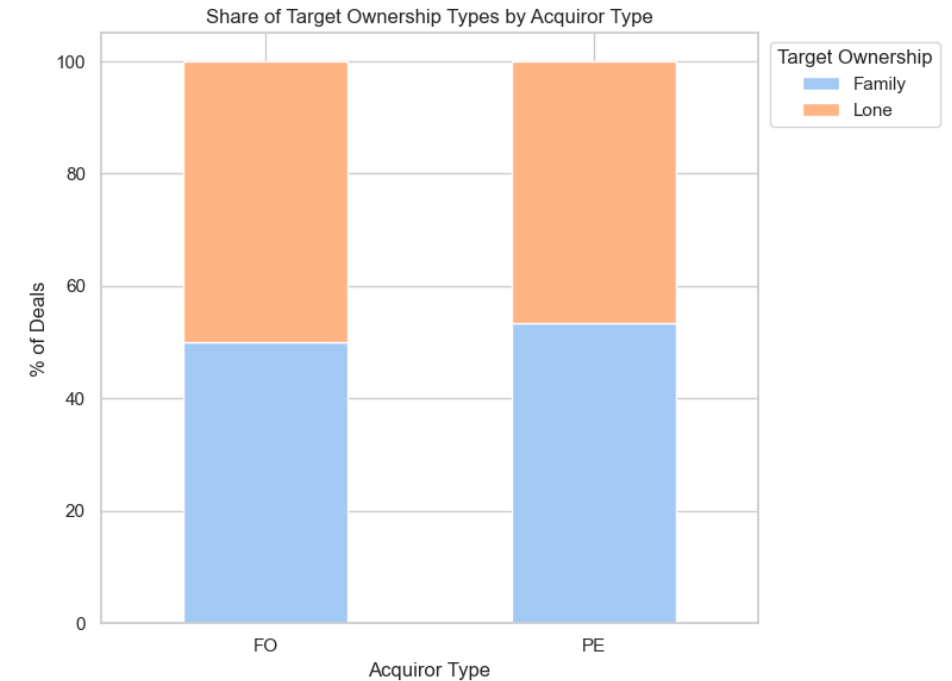
H1 - Investor Preference for Family-Owned Targets

Findings

- FO deals in family firms = **50%**
- PE deals in family firms = **53.4%**
- **Chi² = 0.198, p = 0.656** → not significant (n.s.)
- **Logit coef = -0.458 (p = 0.166)** → confirms no systematic preference

Interpretation

- ❑ **H1 not supported** → no evidence FOs prefer family-owned firms
- ❑ Likely reason:
 - **Heterogeneity & professionalization** of FOs → some resemble PE funds, others legacy-oriented.
 - **Deal characteristics** (size, sector, growth prospects) weigh more than ownership type.
 - **Binary family firm variable** → misses governance, generational stage, and family involvement.



	coef	std err	z	P> z	[0.025	0.975]
const	0.1144	0.636	0.180	0.857	-1.132	1.361
IS_FO	-0.4581	0.330	-1.387	0.166	-1.106	0.189
log_Gearing_t-1	-0.0870	0.125	-0.698	0.485	-0.331	0.157
log_Revenue_t-1	0.0942	0.091	1.032	0.302	-0.085	0.273
wins_ROA_t-1	-0.0303	0.014	-2.112	0.035	-0.058	-0.002
Macro_Finance	2.5713	1.142	2.252	0.024	0.333	4.810
Macro_Industrial	0.9000	0.421	2.138	0.032	0.075	1.725
Macro_Real Estate	-0.8238	1.241	-0.664	0.507	-3.256	1.608
Macro_Services	-0.1482	0.510	-0.291	0.771	-1.147	0.851
Macro_Other	-1.0251	0.675	-1.520	0.129	-2.347	0.297

H2 - Ownership Similarity and Post-Deal Performance

Findings (Robust regression)

- **OWN_SIMILARITY** → +0.214 (p=0.861) → n.s.
- Controls: **log_Gearing(t-1)** + (p≈0.01); **Real Estate sector** + (p≈0.04).

After selection correction (Heckman 2-step)

- **Mills ratio significant** → selection bias exists.
- **OWN_SIMILARITY** → -1.44 (p=0.245) → still n.s.
- Patterns for **leverage** + and **Real Estate** + persist.

	coef	std err	z	P> z	[0.025	0.975]
const	-13.1060	2.380	-5.506	0.000	-17.771	-8.441
OWN_SIMILARITY	0.2136	1.224	0.175	0.861	-2.185	2.613
IS_FAMILY_TARGET	1.3452	1.275	1.055	0.291	-1.154	3.844
log_Gearing_t-1	1.1767	0.469	2.511	0.012	0.258	2.095
log_Revenue_t-1	0.5603	0.352	1.590	0.112	-0.130	1.251
Macro_Finance	3.1511	3.366	0.936	0.349	-3.447	9.749
Macro_Industrial	0.9363	1.750	0.535	0.593	-2.493	4.366
Macro_Real Estate	9.1274	4.472	2.041	0.041	0.362	17.892
Macro_Services	2.5353	2.155	1.176	0.239	-1.689	6.759
Macro_Other	1.2685	2.613	0.485	0.627	-3.853	6.390

	coef	std err	z	P> z	[0.025	0.975]
const	-12.0718	2.375	-5.083	0.000	-16.727	-7.417
OWN_SIMILARITY	-1.4374	1.237	-1.162	0.245	-3.862	0.987
IS_FAMILY_TARGET	0.6424	1.273	0.505	0.614	-1.852	3.137
log_Gearing_t-1	1.5689	0.468	3.351	0.001	0.651	2.487
log_Revenue_t-1	0.3737	0.352	1.063	0.288	-0.315	1.063
Macro_Finance	1.8354	3.360	0.546	0.585	-4.750	8.421
Macro_Industrial	0.4513	1.746	0.258	0.796	-2.972	3.874
Macro_Real Estate	8.1740	4.539	1.801	0.072	-0.723	17.071
Macro_Services	1.8919	2.151	0.879	0.379	-2.325	6.108
Macro_Other	2.0667	2.607	0.793	0.428	-3.043	7.176
mills_ratio	3.1042	0.779	3.984	0.000	1.577	4.631

Interpretation

- ❑ **H2 not supported:** ownership similarity **does not improve short-term ΔROA**.
- ❑ Likely reasons:
 - **Short horizon (1 year):** integration/restructuring costs dominate early.
 - **Metric limits:** ROA may miss FO value creation (governance stability, risk reduction).
 - **Heterogeneity:** FO archetypes and family-firm diversity dilute average effects.

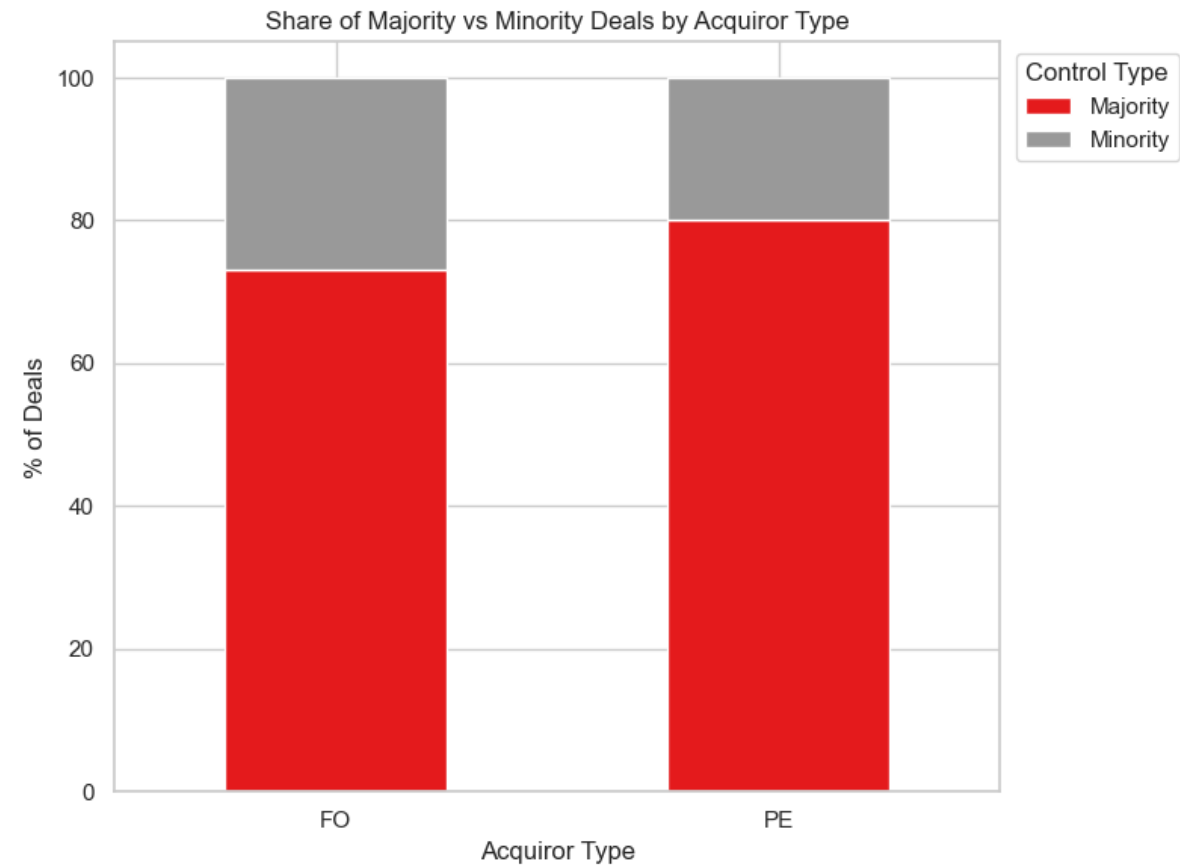
H3 - Deal Type and Investor Identity

Findings

- FO minority stakes = 26.9%
- PE minority stakes = 19.8%
- Majority deals dominate for both: FO = 73.1%, PE = 80.2%
- $\chi^2 = 1.64$, $p = 0.200$ → difference n.s.

Interpretation

- ❑ **H3 not supported** → no significant association between investor type and deal control structure
- ❑ Directionally consistent with literature
- ❑ Likely Reason:
 - **Sample imbalance:** >70% of all deals are majority stakes → limited variation → low statistical power
 - **Strategic differences exist** but may be masked by dataset composition



Key Takeaways

H1 – Investor Preference for Family-Owned Targets

✗ **Not supported** → Family Offices **do not show greater preference** than PEs for family-owned firms.
Investor identity alone does not drive target ownership choice.

H2 – Ownership Similarity and Post-Deal Performance

✗ **Not supported** → Ownership similarity **does not improve short-term performance** (ΔROA_{t+1}).
Potential benefits may emerge only in the long term or in qualitative dimensions (governance, stability).

H3 – Deal Type and Investor Identity

✗ **Not supported** → FOs slightly more open to minority stakes, but **difference not significant**.
Descriptive evidence aligns with theory, but dataset imbalance masks effects.

Discussion & Conclusions

05

Theoretical Contributions



✓ One of the **few quantitative studies** on Family Offices (hand-collected dataset + classification procedure + econometrics).



✓ Offers a **direct FO-PE comparison**, advancing the view of FOs as **hybrid investors** between family firms and PEs.



✓ **Operationalizes ownership similarity** and tests its effects → highlights boundary conditions.

Practical & Managerial Implications



✓ **Family Offices:** ownership similarity doesn't yield short-term gains → effects likely **long-term**, requiring **gradual integration**.

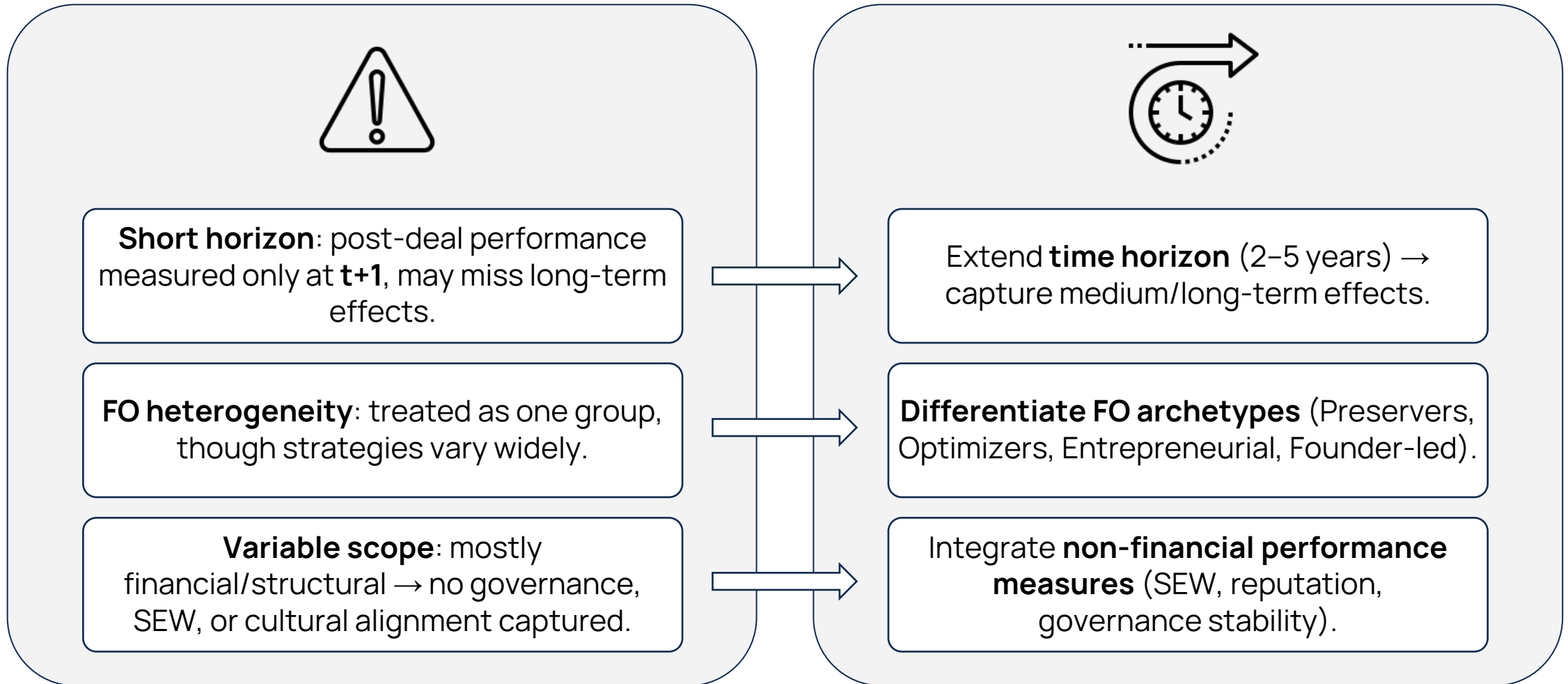


✓ **Private Equity:** results driven by **sector and fundamentals**, confirming focus on **control** over alignment.



✓ **Family Businesses:** “natural partner” perception of FOs may not always reflect reality → assess **strategy, governance** fit.

Limitations & Future Research





Ownership similarity strengthens mutual understanding and governance alignment , its benefits may not appear immediately in financial performance but emerge through **stability, collaboration, and long-term growth.**

Thank you for your attention

Stefano Sima

Questions?

Stefano Sima

Appendix

06

Key Concepts

Family Offices (FOs)

Private entities managing family wealth with a **long-term, value-preserving**, and **relationship-based** approach.



Long-term capital

Private Equity (PEs)

Professional investors managing **external capital**, focused on **short- to medium-term returns** and **exit strategies**.



Active investors

Family Business (FBs)

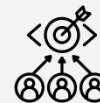
Companies where a **family maintains ownership and control**, guided by **long-term orientation, legacy**, and **continuity**.



Legacy & continuity

Ownership Similarity

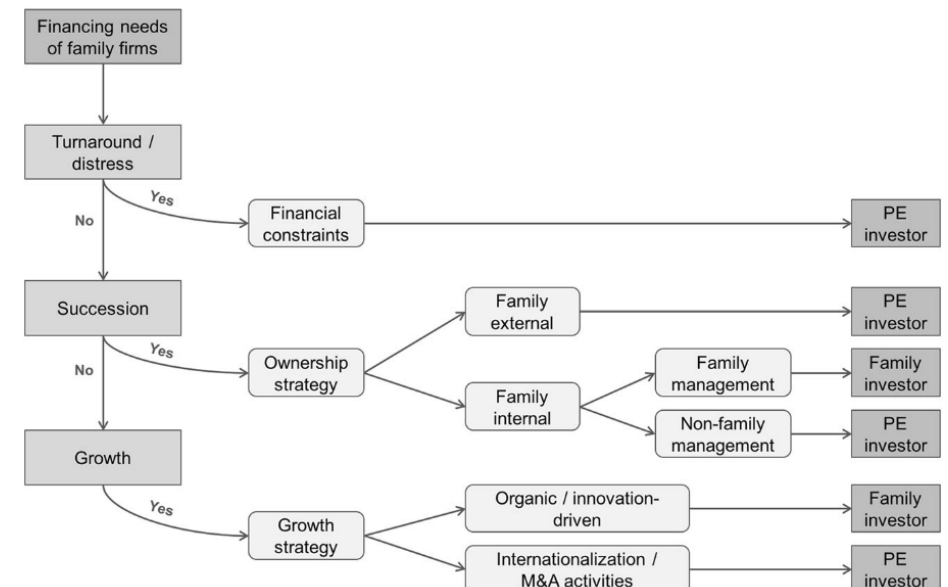
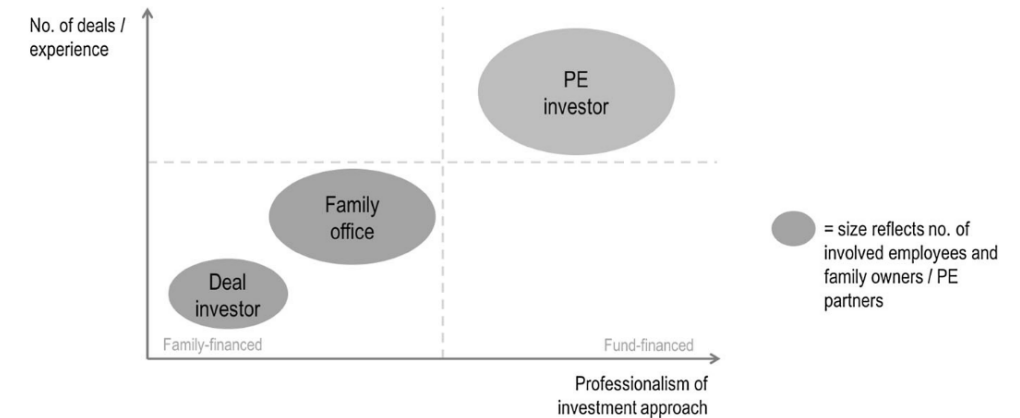
The **alignment** in ownership logic, governance style, and goals between investor and target.



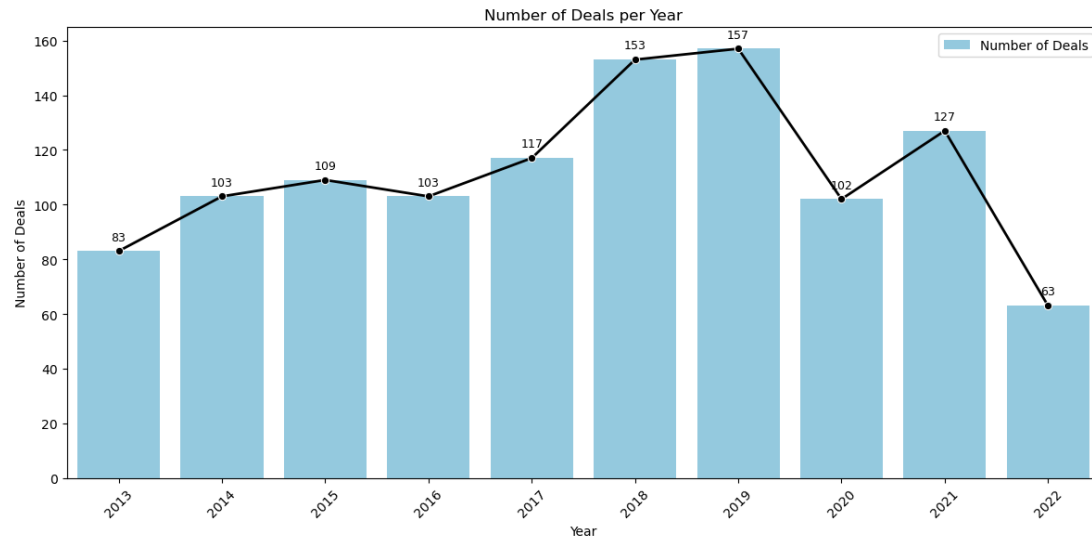
Alignment & fit

Family Offices vs Private Equity

	Family Office	Private Equity
Capital Source	Family wealth (internal capital)	External investors (LPs, debt)
Investment Horizon	Long-term / indefinite	~ 10 years fund cycle
Primary Goal	Preservation and sustainable growth	Maximization of IRR / exit value
Governance Logic	Trust-based, informal, value-driven	Formal, KPI-driven, performance-oriented
Risk Appetite	Moderate / patient capital	High / leveraged
Fit with FB	High	Moderate to low



Exploratory Analysis

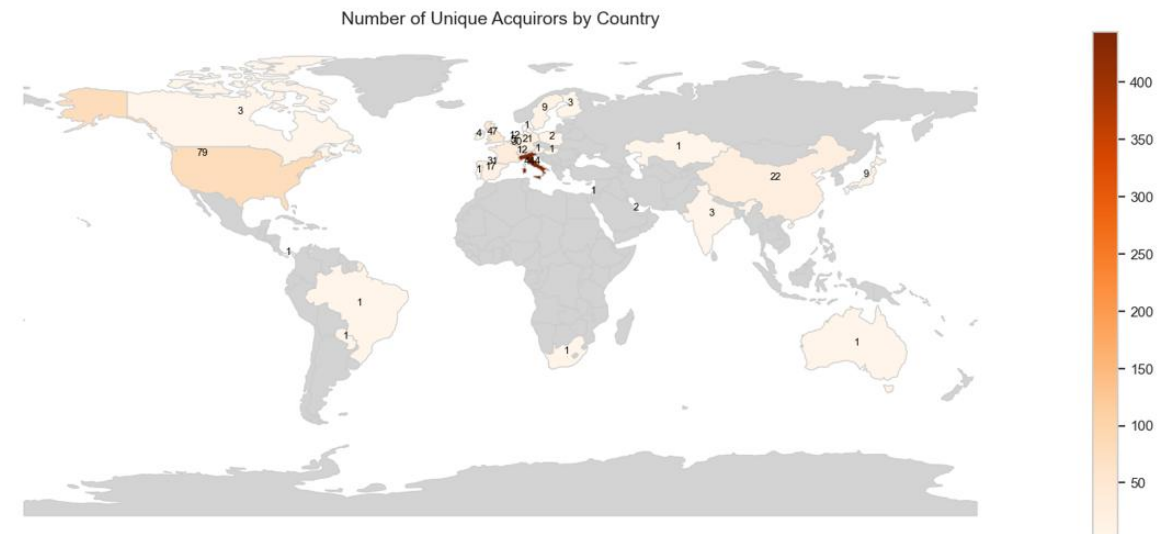


Deal Dynamics

- Transactions grew steadily 2013–2019, peaking at **157 deals in 2019**.
- Sharp decline in **2020** (102 deals) → COVID-19 impact.
- **Partial rebound in 2021** (127 deals), drop again in 2022 (63 deals).
- Dataset captures both **expansion and slowdown** phases.

Geography of Acquirors

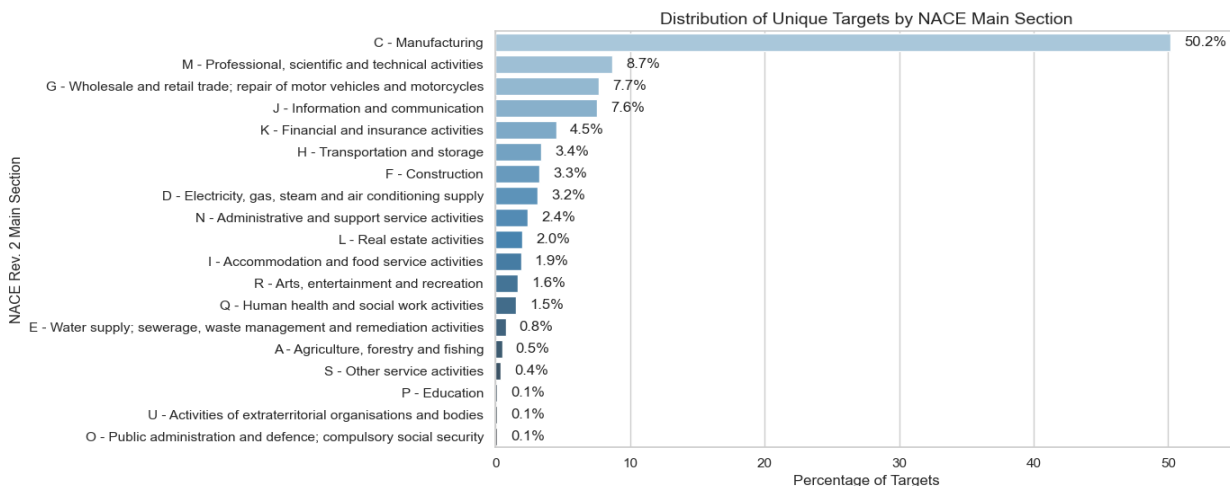
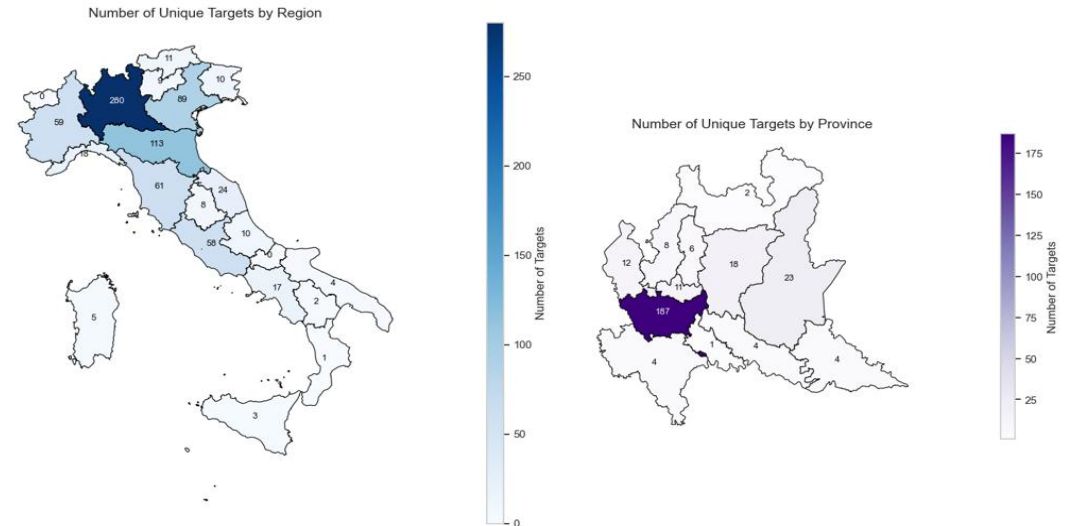
- **Italy = core market** with 444 unique acquirors → expected, since **all targets are Italian**.
- Strong cross-border presence from US (79), UK (47), France (31), Luxembourg (30), China (22), Germany (21).
- Pattern shows Italy as both a **domestic hub** and an **international arena** for corporate transactions.



Exploratory Analysis

Geography of Targets

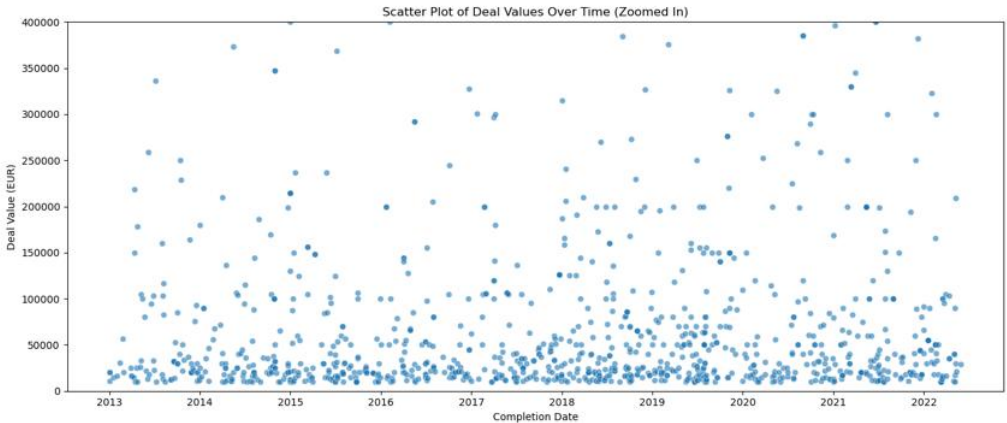
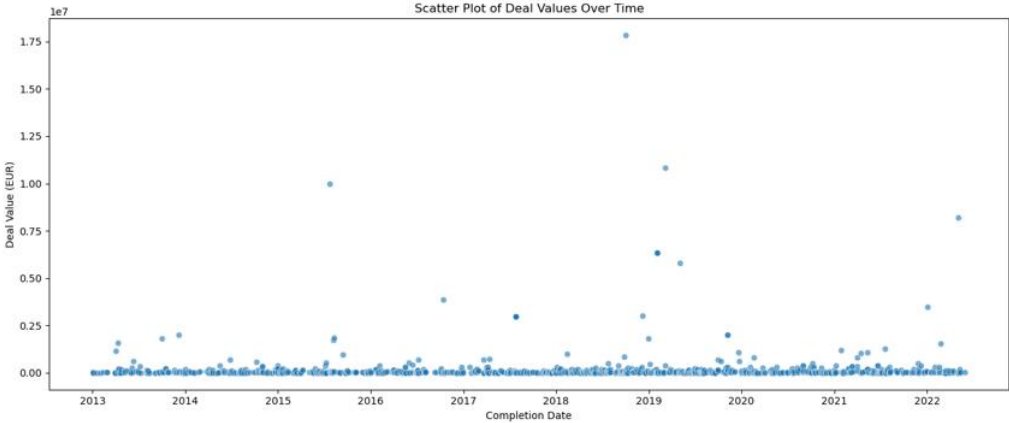
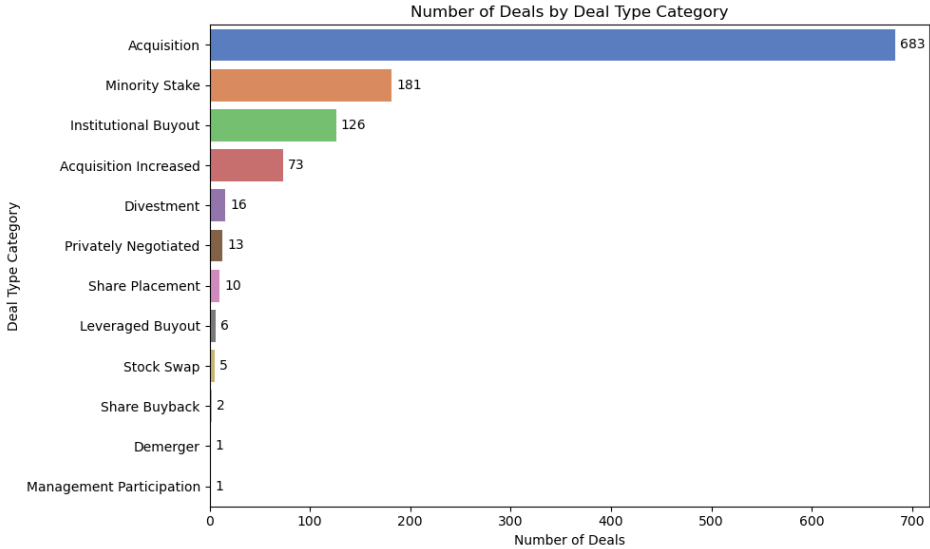
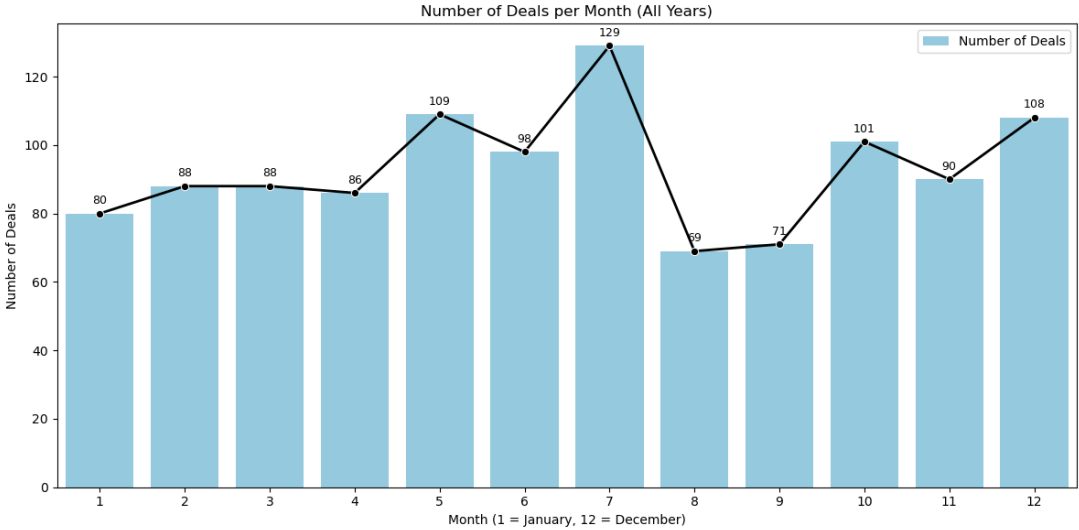
- Strong **northern concentration**: Lombardy 280 (Milan 187), Emilia-Romagna 113, Veneto 89.
- Reflects Italy's economic geography: **industrial north = core of investment activity**.
- Milan stands out as Italy's **financial & business hub**, attracting both domestic and foreign deals.
- Limited deal presence in the south → highlights **territorial asymmetries**.



Sectors of Targets

- **Manufacturing = 50%** of targets → backbone of Italian economy.
- Other key sectors: **Professional & technical (8.7%)**, **Wholesale & retail (7.7%)**, **ICT (7.6%)**.
- Lower share in finance (4.5%) & real estate (2.0%) → deals **not dominated by speculative logic**.
- Broad diversification across 20 sectors → reflects **strategic, long-term orientation** of investors.

Exploratory Analysis (Deals)



Post-hoc Power Analysis (G*Power)

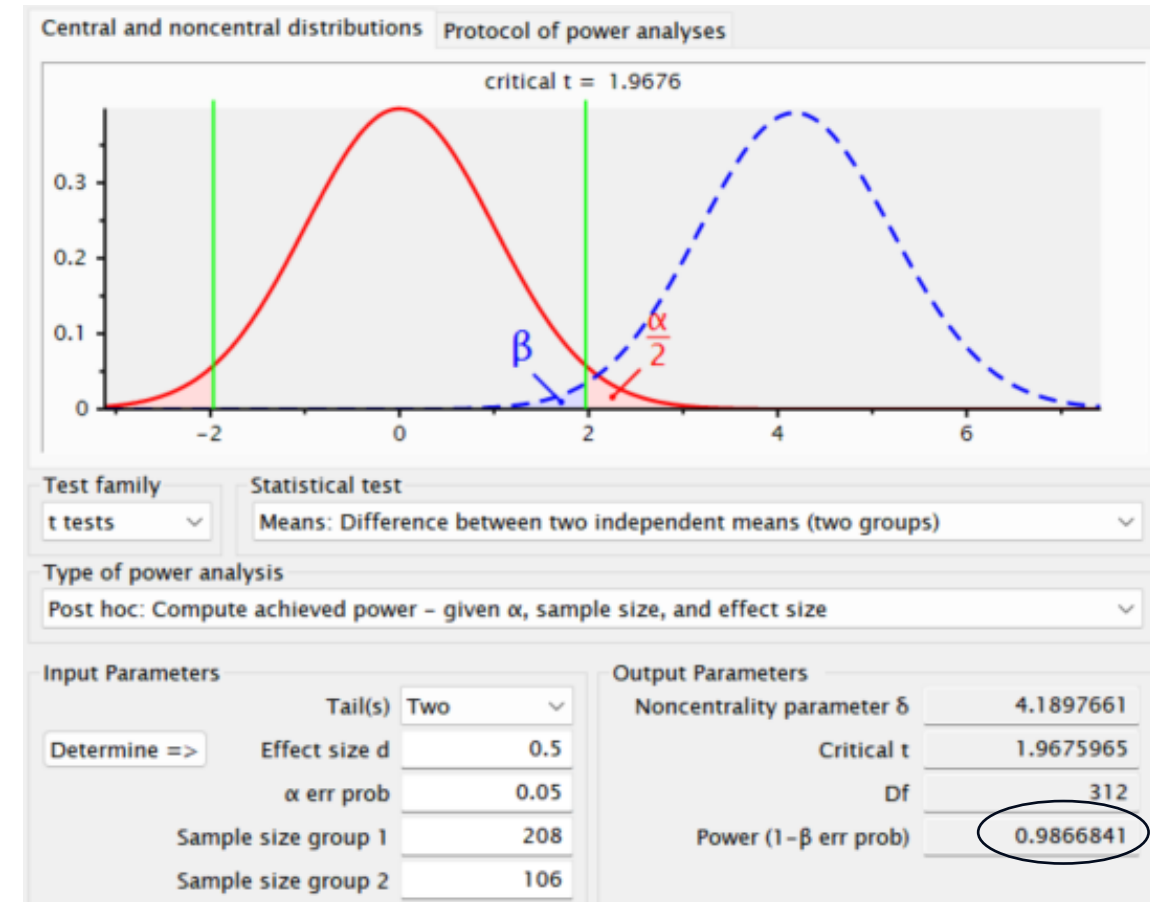
Purpose

To verify whether the available sample size provides adequate **statistical power** for detecting meaningful differences between **Family Offices** and **Private Equity** acquirors.

Configuration

- **Test family:** *t* tests (difference between two independent means)
- **Tail(s):** Two-tailed
- **Effect size (Cohen's d):** 0.5 (medium effect, Cohen 1988)
- **Significance level (α):** 0.05
- **Sample sizes:** $n_1 = 208$ (PE), $n_2 = 106$ (FO)

The study is **well-powered** (power > 0.80) to detect medium effects, confirming that non-significant results are **unlikely due to insufficient sample size (Type II error)**.



Key Variables Constructed (1/2)

To test **H1–H3**, several new analytical variables were created to capture investor type, target characteristics, and ownership alignment.

Investor & Target Type

- **IS_FO** = 1 if acquiror is a *Family Office*, 0 if *Private Equity*.
- **IS_FAMILY_TARGET** = 1 if target is *family-owned* pre-deal.

Ownership Similarity

- = 1 when **FO** → **Family firm** or **PE** → **Lone firm** (aligned ownership models).
- = 0 otherwise.

Interaction Terms

- **OWN_SIM_X_FAMILY** → Effect of similarity among family targets.
- **FO_X_FAMILY** → Specific effect of FO acquiring a family firm.

Key Variables Constructed (2/2)

To test **H1–H3**, several new analytical variables were created to capture investor type, target characteristics, and ownership alignment.

Sector Dummies

- Aggregated NACE codes into **5 macro-sectors**: Industrial, Finance, Real Estate, Services, Other.
→ Avoids overfitting and ensures more stable estimates.

Control Type

- **Majority deals**: full acquisitions, buyouts, acquisition increases.
- **Minority deals**: share placements, stock swaps, negotiated investments.
→ Used to test **H3** on differences in control preferences.

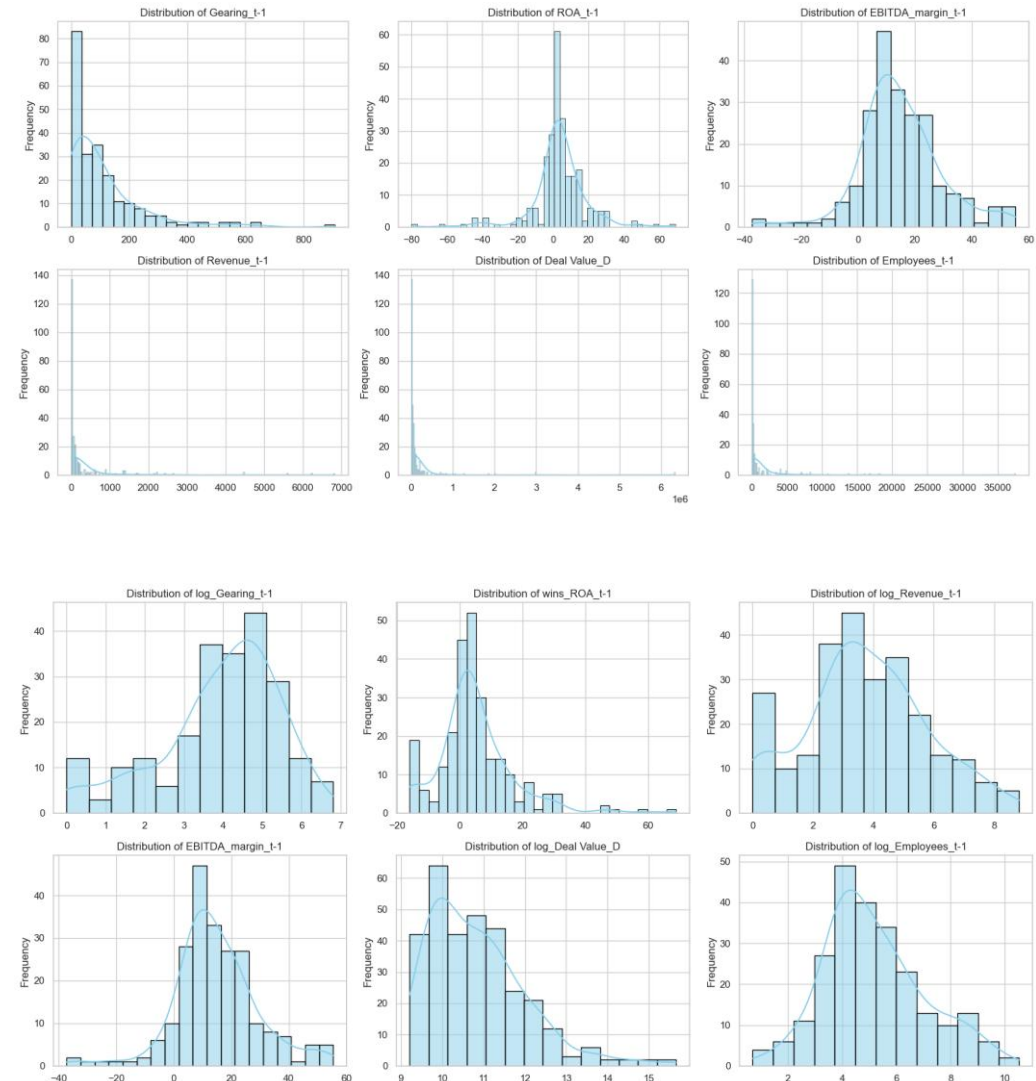
Variable Transformations

To address **skewness** and **outliers** in firm-level data, two transformations were applied:

- **Logarithmic transformation** (*Gearing, Revenue, Employees, Deal Value*): reduced extreme values and improved normality.
- **Winsorization** (*ROA*): applied asymmetrically (5th–1st percentiles) to limit outlier effects while keeping all observations.



These steps made variables **more regular** and **robust for estimation**.



Heckman Two-Step Approach

To address potential **selection bias**, since firms acquired by Family Offices may not be randomly selected, a **Heckman correction** was applied.

Step 1 – Selection Equation (Probit Model)

Estimates the probability of being acquired by a **Family Office**:

→ Identifies factors influencing **FO vs PE investment choice**.

$$\Pr(FO_i = 1) = \Phi(\beta_0 + \beta_1 IS_FAMILY_TARGET_i + \beta_2 \log(Revenue_{t-1,i}) + \beta_3 \log(Gearing_{t-1,i}) + \sum_{s=1}^5 \beta_{3+s} Macro_Sector_{s,i})$$

Inverse Mills Ratio (λ_i)

Derived from the predicted probabilities:

→ Captures **latent selection effects** influencing deal type.

$$\lambda_i = \frac{\phi(\hat{p}_i)}{\Phi(\hat{p}_i)} \quad \text{if } IS_FO = 1, \quad \text{or} \quad \lambda_i = \frac{-\phi(\hat{p}_i)}{1 - \Phi(\hat{p}_i)} \quad \text{if } IS_FO = 0$$

Step 2 – Outcome Equation (Performance Model)

Includes λ_i to correct bias when estimating **post-deal performance (ΔROA)**:

$$\Delta ROA_i = \beta_0 + \beta_1 OWN_SIMILARITY_i + \beta_2 IS_FAMILY_TARGET_i + \beta_3 \log(Gearing_{t-1,i}) + \beta_4 \log(Revenue_{t-1,i}) + \sum_{s=1}^5 \beta_{4+s} Macro_Sector_{s,i} + \beta_{10} \lambda_i + \varepsilon_i$$

Robustness Checks and Model Validation (1/2)

Objective

Validate the **robust regression** and **Heckman two-step model** used to test whether **ownership similarity** affects post-deal performance.

Key Diagnostics

Breusch-Pagan test:

- $p = 0.957 \rightarrow$ constant variance (homoscedasticity confirmed)

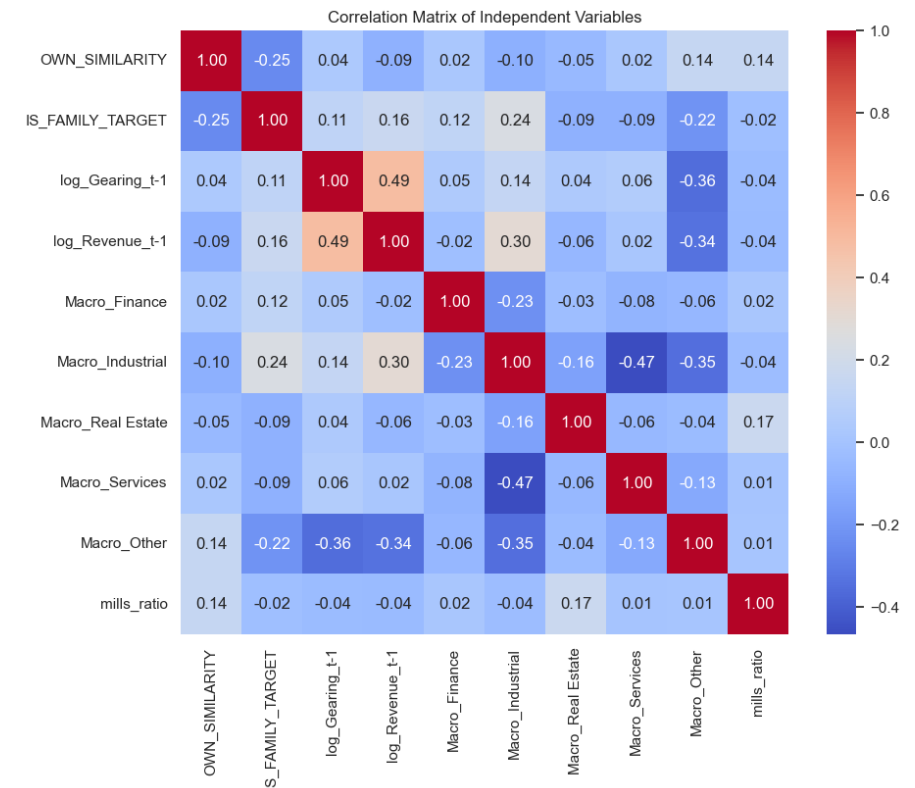
Selection correction:

- **Significant Mills ratio (λ)** $\rightarrow$ selection bias properly addressed

Multicollinearity:

- All VIFs $< 10 \rightarrow$ acceptable levels
- Highest values: $\log_Gearing_{t-1} = 8.4$, $\log_Revenue_{t-1} = 6.7$
- Max correlation = 0.49 $\rightarrow$ no serious collinearity

Feature	VIF	Feature	VIF
OWN_SIMILARITY	1.918	Macro_Industrial	4.169
IS_FAMILY_TARGET	2.634	Macro_Real Estate	1.127
$\log_Gearing_{t-1}$	8.446	Macro_Services	1.684
$\log_Revenue_{t-1}$	6.704	Macro_Other	1.199
Macro_Finance	1.242	mills_ratio	1.062



Robustness Checks and Model Validation (2/2)

Key Diagnostics

Robust weights:

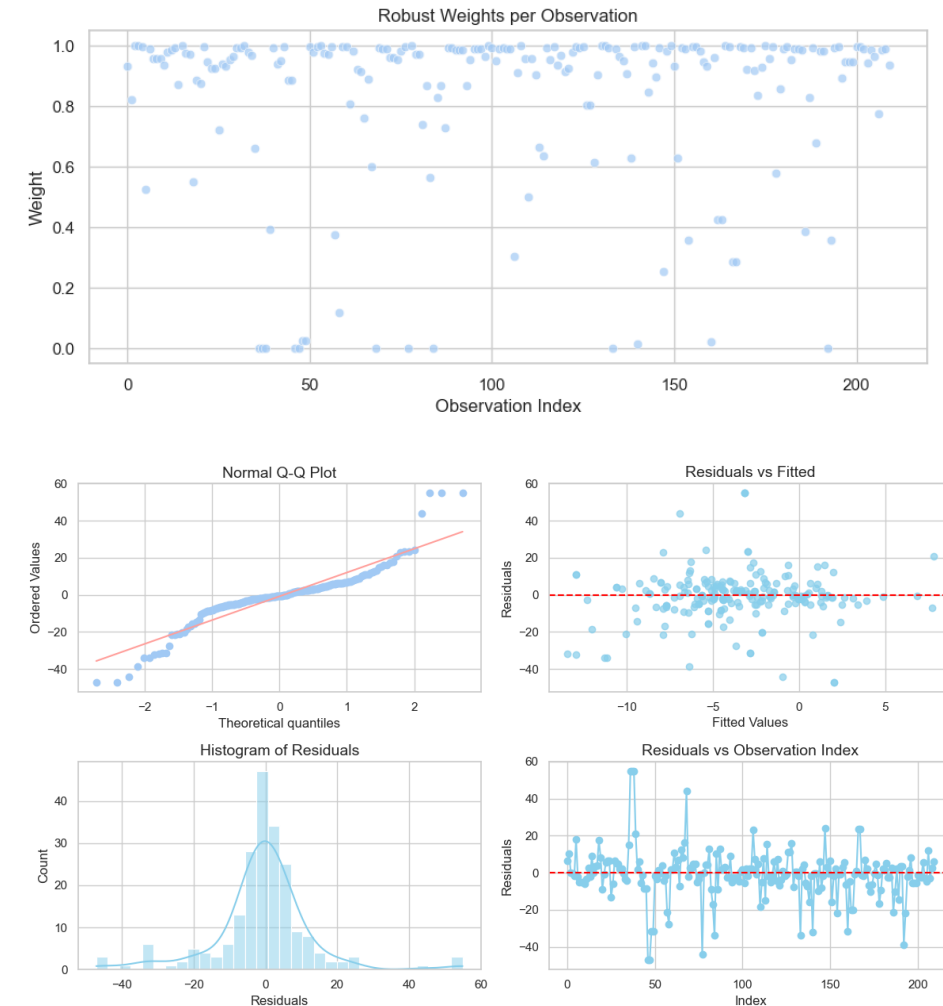
- Most observations with weight $\approx 1 \rightarrow$ limited impact of outliers
- Few low-weight cases $\rightarrow$ influence successfully reduced

Residual analysis:

- Residuals centered around zero $\rightarrow$ no visible patterns
- Slight right-skewness but acceptable under robust estimation
- No heteroscedasticity or autocorrelation detected

The **Heckman model** and **robust regression** meet all key statistical assumptions:

- ✓ coefficients are stable and reliable
- ✓ outliers and heteroscedasticity effectively controlled
- ✓ no multicollinearity or bias distortions



Heterogeneity within Family Offices (1/2)

Objective

Provide a **preliminary exploration** of potential **differences between subgroups of Family Offices**:

- **Financialized FOs** → families that divested their original business and act mainly as financial investors.
- **Co-existing Core-Business FOs** → families still directly linked to an operating family enterprise.

Aim: identify whether these subgroups differ in:

1. **Ownership preference** (Family vs Non-family targets)
2. **Deal structure** (Majority vs Minority stakes)

Methodology Overview

- Dataset filtered to include only **FO transactions (N = 106)** → 53 per subgroup.
- Created dummy variable **FO_Financialized = 1** if the FO has no active family business.
- Replicated main thesis methodology (as in H1 & H3): **Chi-squared tests, logistic regression, and descriptive comparisons.**
- Controls included: *leverage, size, profitability, macro-sector dummies.*

**Achieved power
=0.72 (< 0.80
threshold)**

*Exploratory result,
moderate risk of
Type II error (false
negatives).*

Heterogeneity within Family Offices (2/2)

	Co-existing Core-Business FO	Financialized FO	Difference	Significance
Family-owned targets	47.2%	52.8%	+5.6 pp	$\chi^2 = 0.15$ ($p = 0.698$)
Minority deals	19.6%	34.0%	+14.4 pp	$\chi^2 = 2.04$ ($p = 0.153$)

Key takeaways

- Financialized FOs show *slightly* higher share of family-owned targets and minority stakes.
- Differences are **not statistically significant** ($p > 0.10$).
- Results reflect **descriptive variation**, not systematic divergence.

Interpretation

- Analysis is **descriptive**, not inferentially conclusive.
- Results should be read as **preliminary insights** into intra-FO heterogeneity.
- Serves as a **basis for future research** with larger samples and refined typologies.

